



DRISHTI

Finding missing people in real time at the Kumbh Mela

A real-time face-matching prototype for the Nashik-Trimbakeshwar Simhastha Kumbh Mela 2027 — designed for a crowd of ~12 crore pilgrims.

Detect → **Embed** → **Match** → **Reunite**

THE PROBLEM

Getting lost in the world's largest gathering

The Kumbh Mela is the biggest peaceful human gathering on Earth — and separation from family is one of its most common, frightening experiences.

The traditional fix — a loudspeaker and a lost-and-found (Bhula-Bhatka) tent — is slow, passive, and useless for someone who is disoriented, wandering, or can't ask for help.

The elderly, children, and non-local-language speakers are most at risk, and every hour of separation is acute distress.

Why it's hard

- Tens of thousands reported missing across a single mela
- Up to 2.5 crore people on a single peak day
- A wandering person rarely stays near a booth
- Manual CCTV review can't keep up with the crowd
- Children change appearance fast & have few photos

We design for the actual deployment

12 cr

Pilgrims expected

across the full mela

2.5 cr

Peak-day footfall

≈ 25 million in one day

4,011

AI CCTV cameras

+ 18 surveillance drones

₹300 cr

Surveillance budget

state "Kumbh AI Stack"

Where the cameras are — and where people get lost

Ramkund ghats (densest zone) · surrounding ghats · Sadhugram akhada camps · Trimbakeshwar temple (~30 km away) · Nashik Road railway station · CBS bus stand · Ramkund Marg corridor.

Two sites, 30 km apart, one search space — a person lost at Ramkund may resurface at Trimbakeshwar. That's why DRISHTI uses ONE central vector index across all 4,011 cameras, not per-camera search.

IT ALREADY WORKS

Prayagraj 2025 proved the concept

At the Maha Kumbh 2025, authorities used AI facial recognition to locate lost people for the first time — when someone was reported missing, the system scanned the camera network and traced their route through CCTV. India's police already run a national face-recognition system that has matched thousands of missing-child cases.

2,750+

AI CCTV cameras

Prayagraj 2025

100

Face-recog cameras

at major stations

10

Lost-&-found booths

tech-equipped

1000s

Missing children

matched by AFRS

DRISHTI takes this proven idea and purpose-builds it for Nashik 2027: family-facing, privacy-bounded, and tuned to the site's geography.

THE SOLUTION

DRISHTI: an active search layer

Family gives a photo → DRISHTI searches the live camera network → reports where that face appears, in near-real time.

1

Register

Family submits a photo + consent at a booth or in the app. DRISHTI stores only a face-print.

2

Search

Every face on every camera is matched against the watch-list in milliseconds.

3

Locate

A match pins the camera, location & time and reconstructs the person's path.

4

Reunite

Family + nearest booth are alerted; an operator confirms before dispatch.

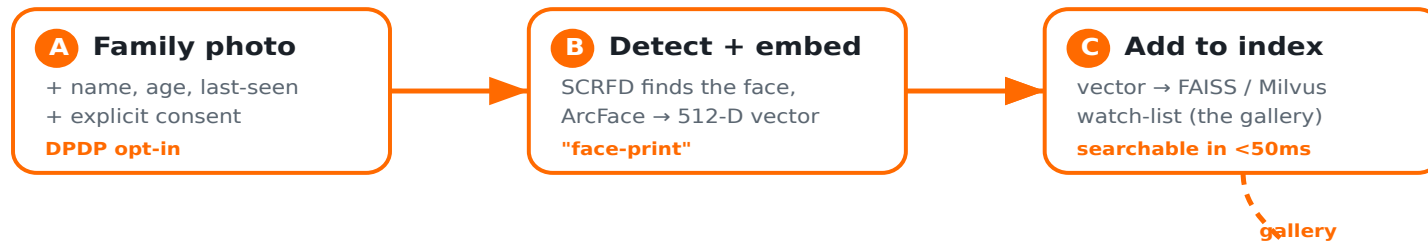
Opt-in reunification — not blanket surveillance.

HOW IT WORKS

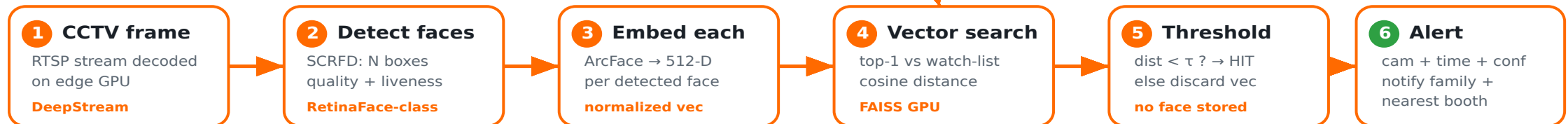
Detect → Embed → Match → Alert

DRISHTI — Real-Time Matching Pipeline (per frame)

ENROLLMENT (once, when a person is reported missing)



LIVE INFERENCE (every frame, every camera, 24×7)



Cross-camera continuity & safeguards

- **ByteTrack** links the same person across consecutive frames and neighbouring cameras → reconstructs the person's path through the mela.
- **Human-in-the-loop:** a control-room operator confirms each high-confidence hit before the family is dispatched — the model assists, people decide.
- **Privacy by design:** non-matching faces are never stored; only watch-list (consented) embeddings persist, and all are purged when the mela ends.

Edge · Core · Consumers

DRISHTI — End-to-End System Architecture

Nashik Kumbh Mela 2027 · Missing Person Finder



Proven open-source, built to scale

Detect faces

SCRFD / RetinaFace

strong even in dense, occluded crowds

Embed (face-print)

ArcFace • InsightFace

512-D vector, robust to light & pose

Match at scale

FAISS / Milvus

top-k over millions of vectors in <50 ms

Track across cameras

ByteTrack

reconstructs the person's route

Run 4,011 streams

NVIDIA DeepStream + TensorRT

edge GPU; sends vectors, not raw video

≈100 edge GPUs feed one central vector index — comfortably within the ₹300 cr envelope.

A working demo you can run today

index.html — zero install

- Real face recognition in the browser (TensorFlow.js)
- Register a missing person from a photo
- 12-tile simulated Kumbh camera wall
- Live detection, matching, boxes & alerts
- Match log + command dashboard

backend_reference.py — production stack

- Same logic, real models: SCRFD + ArcFace + FAISS
- FastAPI: /enroll · /search · /watchlist · /stats
- Interactive Swagger docs
- GPU-ready: swap to onnxruntime-gpu + faiss-gpu
- The blueprint for the real deployment

Demo flow: register a sample face → drop a crowd photo into a feed → watch DRISHTI find them.

Built for the DPDP Act 2023

1 Consent & purpose limit

Matches ONLY against photos a family explicitly submitted to find a specific person.

2 Data minimisation

Non-matching faces are never stored. Raw video stays at the edge — only vectors travel.

3 Automatic deletion

All watch-list embeddings & logs purged when the mela ends.

4 Human oversight

Every match is operator-confirmed; every access is logged & auditable.

Civil-society groups warn against weak guardrails. We treat those concerns as design requirements — governance is a first-class layer, not an afterthought.

Measured in reunions, not faces scanned

Success metrics

- Median time-to-reunion (minutes, not hours)
- % of reported missing persons located
- Children located — highest priority
- False-positive rate at operating threshold
- Accuracy parity across demographics

Roadmap

- **Prototype — demo + reference backend ✓**
- Pilot — Ramkund ghat cluster, one booth
- Scale-out — all 4,011 cameras + family app
- Harden — bias audit, DPDP layer, peak load
- Reuse — future melas + stampede warning



Bring families back together — in real time, at any scale.

DRISHTI • Kumbh Mela Missing Person Finder

Prototype + research report + architecture + this deck — all in one package.